Efficiency Consequences of Visit Schedule Choice in Longitudinal Trials: An Analytic and Simulation-Based Comparison

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Abstract

Observation times in longitudinal clinical trials are usually spaced equally for administrative convenience, but optimal design theory shows that placement affects the precision of estimated treatment effects. We compare an equally spaced five-visit schedule with a boundary-clustered schedule for a two-arm, 24-month trial with a linearly divergent treatment effect, combining an exact generalized least squares (GLS) efficiency calculation with a common-random-number Monte Carlo study (2000 paired replications) under a random-intercept, random-slope data generating model. The analytic relative efficiency of the clustered design is 1.06, and the paired simulation, after recalibrating the effect size so power is informative rather than at ceiling, reproduces this result (empirical relative efficiency 1.07, 95% bootstrap interval 1.06-1.09), confirming that the precision advantage is real and not simulation noise. A subsequent Monte Carlo grid crossing five covariance structures with four visit schedules corroborates the direction of this finding in 14 of 15 non-baseline cells, with the advantage shrinking as the random-slope variance grows large relative to the residual variance. We do not claim optimality: the theoretically optimal design for this covariance structure is not computed here, and the primary paired comparison uses only two designs. The contribution is a worked, fully reproducible demonstration of how much efficiency is at stake for one realistic configuration, and a template (`R/simulation.R`) for extending the comparison to dropout mechanisms and further candidate schedules.

Contents

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1 Introduction

In longitudinal clinical trials, subjects are measured repeatedly over a fixed study period. The conventional approach assigns observation times at equally spaced intervals (monthly or quarterly visits, for example) largely for administrative convenience rather than statistical efficiency. A growing body of work in optimal experimental design suggests that the placement of these observation times can substantially affect the precision with which treatment effects are estimated. We begin in Section 1.1 with a review of the relevant literature and then describe the analytic efficiency calculation and simulation study, and their results, in Sections 2 and 3.

1.1 Optimal allocation of observation times

[1] provided the foundational analysis of this problem for two-group trials with linearly divergent treatment effects. They showed that the optimal allocation of time-points depends critically on the assumed covariance structure. Under compound symmetry (CS), efficiency improves by clustering repeated measures near the end of the study period, where the treatment effect is largest. Under a first-order autoregressive (AR(1)) structure, only two measurements – at baseline and at study end – are needed for highly efficient estimation. Under a more realistic AR(1) plus measurement error structure, the optimal spacing depends on the specific covariance parameter values, but the equally spaced design is generally suboptimal.

In a companion paper,[2] extended these results to address the joint optimization of the number of repeated measures and group sizes under budget constraints. They found that when budget is fixed and the covariance structure includes an autoregressive component, two repeated measures per subject often yield highly efficient treatment effect estimators. Otherwise, intermediate measures can reduce the variance of the generalized least squares (GLS) estimator, particularly under compound symmetry.

1.2 Design criteria and cost considerations

The formal framework for these results relies on classical optimal design theory, as developed by[3].[4] proposed maximin D-optimal designs for longitudinal mixed effects models, computing optimal allocations of time-points for polynomial trend models with random intercepts, random slopes, and AR(1) serial correlations. Because D-optimal designs depend on unknown parameter values, the maximin criterion provides robustness by maximizing the worst-case relative efficiency across the parameter space.

[5] extended this work to cohort designs, showing that a purely longitudinal design with a single cohort measured at optimal time-points is generally more efficient than mixed longitudinal-cross-sectional designs. Their numerical results demonstrated that the most efficient number of repeated measurements equals the sum of the number of cohorts and the degree of the polynomial trend. Subsequently,[6] showed that adding cohorts or repeated measurements beyond this optimum wastes resources when the study budget is fixed.

1.3 Practical guidance on the number of measures

From a more pragmatic perspective,[7] examined the marginal benefit of adding repeated measures in terms of statistical power. He showed that while repeating assessments can have dramatic effects on power, the marginal benefit of each additional measure decreases rapidly. In general, there is little value in exceeding four post-treatment assessments (or seven total including baseline), except when correlations between measures are low, as in episodic conditions.

1.4 Extensions and related work

Several extensions of the Winkens framework have appeared.[8] generalized the optimal design results to second-order polynomial treatment effects, finding that equidistant time-points are either optimal or highly efficient when only three measures are taken.[9] investigated optimal designs when the goal is to estimate individual growth curves (random effects) rather than fixed effects, showing that the optimal number of time-points and subjects differs from the fixed-effects case.[10] developed Bayesian D-optimal designs for dichotomous repeated measures with autocorrelation, formally accounting for parameter uncertainty through prior distributions.

For the analysis of unequally spaced longitudinal data,[11] provided the statistical machinery for fitting mixed models with continuous-time AR(1) structures and arbitrary observation schedules. Their approach, based on the Kalman filter, remains foundational for analyzing data from designs with non-equidistant time-points.

1.5 Power and sample size tools

Modern software for power and sample size calculations in longitudinal trials has made these design considerations more accessible.[12] describe the `longpower` R package, which implements closed-form power calculations for linear mixed models.[13] applied these methods to Alzheimer's disease Phase II trial design, demonstrating the practical impact of design choices on the ability to detect disease-modifying treatment effects.

1.6 Present study

The theoretical optimality results reviewed above establish that the best design depends on the assumed covariance structure, but they do not by themselves quantify how much efficiency is at stake for a specific, realistic trial configuration, nor do they address the finite-sample behavior of the REML-based inference that practitioners actually use. This report addresses that translational gap for one configuration: a 100-patient, two-arm, 24-month trial with a linearly divergent treatment effect, analyzed under a random-intercept, random-slope model. We compare two five-visit designs:

- **Equal spacing:** observations at months 0, 6, 12, 18, and 24.
- **Clustered spacing:** observations clustered near the beginning and end of the study period (months 0, 2, 4, 20, 24). Under a random-slope covariance structure (Section 2.1), boundary-loaded time-points are efficient for estimating the slope contrast for a classical leverage reason – observations far from the mean time carry more information about a slope – which is distinct from the compound-symmetry mechanism that motivates clustering in[1]; see Section 2.2 for the derivation.

We first derive the exact asymptotic (GLS) relative efficiency of the two designs in closed form, then use a common-random-number Monte Carlo study to assess whether finite-sample REML estimation and inference (bias, empirical versus model-based standard error, power, and 95% confidence interval coverage) reproduce the analytic prediction. We do not claim that either design is optimal: the theoretically optimal design implied by the cited literature is not computed here, and the main paired comparison considers only two candidate schedules, although a Monte Carlo sensitivity grid over four schedules and five covariance structures (Section 2.7) corroborates the direction of the finding more broadly. `R/simulation.R` in the accompanying compendium implements the data-generating model, the analytic efficiency calculation, and the paired simulation as reusable functions so the comparison can be extended to other covariance structures, sample sizes, dropout mechanisms, and candidate schedules (Sections 2.6 and 2.7).

2 Methods

2.1 Data generating model

We simulate data from a linear mixed effects model with random intercepts and random slopes. Let Y_{ij} denote the outcome for subject i at time t_j. The generating model is:

$$Y_{ij} = (\beta_0 + b_{0i}) + (\beta_1 + b_{1i})\, t_j + \beta_2\, G_i + \beta_3\, G_i\, t_j + \varepsilon_{ij}$$

where $G_i \in \{0, 1\}$ is the treatment group indicator, b_{0i} and b_{1i} are subject-specific random effects with $\mathrm{Var}(b_0) = \tau_0^2$, $\mathrm{Var}(b_1) = \tau_1^2$, $\mathrm{Cov}(b_0, b_1) = \tau_{01}$, and $\varepsilon_{ij} \sim N(0, \sigma^2)$.

The parameter of interest is β_3, the treatment-by-time interaction, which represents the rate at which the treatment effect diverges from the control effect per unit time.

2.2 Covariance structure and theoretical prediction

Marginalizing over the random effects, this model induces a within-subject covariance $\mathrm{Cov}(Y_{ij}, Y_{ij'}) = \tau_0^2 + \tau_{01}(t_j + t_{j'}) + \tau_1^2 t_j t_{j'} + \sigma^2 \mathbb{1}(j = j')$, in which the marginal variance increases with t_j^2 and the correlation between visits depends on their product $t_j t_{j'}$, not on their lag $|t_j - t_{j'}|$. This is the standard random-coefficients (random-intercept, random-slope) covariance structure. It is distinct from both compound symmetry (CS), which is the random-intercept-only special case ($\tau_1 = 0$) with constant correlation regardless of visit timing, and from a first-order autoregressive (AR(1)) structure, whose correlation depends only on the lag between visits and does not grow with τ_1 boundary leverage. An earlier draft of this report described the simulated structure as combining features of CS and AR(1); that characterization is incorrect and has been removed (see the Discussion for the corrected interpretation).

The efficiency prediction relevant to this structure differs from the CS-based argument in[1]. Under CS, clustering observations near the end of the study helps because the treatment effect (and hence the signal) is largest there while every pair of visits is equally correlated, so nothing is lost by leaving a gap. Under the random-slope structure simulated here, the reason boundary-clustered designs are more efficient for the *slope* contrast β_3 is instead the classical leverage argument for regression slopes: for a simple linear regression of Y on t, $\mathrm{Var}(\widehat{\text{slope}}) \propto 1/\sum_i (t_i - \bar{t})^2$, so observations placed far from the mean time carry disproportionately more information about the slope. Section 2.5 below verifies this prediction is directionally correct but shows the efficiency gain shrinks as the random-slope variance τ_1^2 grows relative to the residual variance, because a larger τ_1^2 makes the slope itself (not just its estimation error) more variable across subjects, which increases the total variance contributed by every visit and dilutes the boundary-leverage advantage.

2.3 Parameter values

We use the following parameter values, broadly motivated by cognitive decline trials:

Table 1: Simulation parameters.

Parameter	Symbol	Value
Intercept	β_0	50.00
Time slope	β_1	-1.00
Group effect at baseline	β_2	0.00
Group x time	β_3	0.19
SD random intercept	τ_0	5.00
SD random slope	τ_1	0.30
Residual SD	σ	3.00
Subjects per group	n	50.00
Study duration (months)	–	24.00
Number of simulations	–	2000.00

The treatment-by-time interaction $\beta_3 = 0.19$ implies that the treatment group improves (or declines less) by 0.19 units per month relative to control, yielding a cumulative treatment advantage of 4.56 units over 24 months.

2.3.1 Recalibration of the treatment effect

An earlier draft used $\beta_3 = 0.5$, for which the analytic power of both designs is 1 to machine precision (Section 2.5): every replication rejects the null, so power cannot discriminate between designs and the metric is uninformative. We recalibrated β_3 so that the equal-spacing design has approximately 80% analytic power at the two-sided $\alpha = 0.05$ level, which requires $\beta_3/\mathrm{SE}_{\mathrm{equal}} \approx 2.80$ (the sum of the z-quantiles for 80% power and a two-sided 5% test). With $\mathrm{SE}_{\mathrm{equal}} = 0.0678$ (Section 2.5), this gives $\beta_3 \approx 0.19$, for which the analytic power is 80.0% (equal spacing) and 82.4% (clustered spacing) – close to, and not at, the ceiling, so that power remains informative while both designs are still adequately powered for a realistic trial. All other parameters are unchanged from the original design.

2.4 Design specifications

Table 2: Observation times (months) for each design.

Design	Visit 1	Visit 2	Visit 3	Visit 4	Visit 5
Equal	0	6	12	18	24
Clustered	0	2	4	20	24

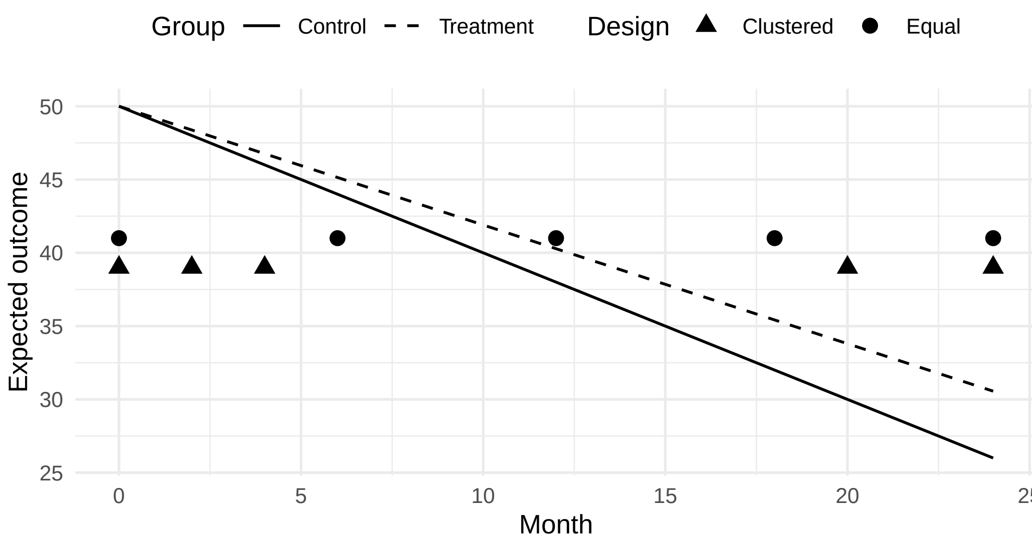

Figure 1: Schematic of the two visit schedules over the 24-month study period. Lines show the expected group trajectories; markers show each design's visit times, vertically offset (Equal above, Clustered below) purely to avoid overplotting at shared visit times (0 and 24 months) and carry no other meaning.

2.5 Analytic relative efficiency

Under the data generating model of Section 2.1, the treatment-by-time interaction $\hat{\beta}_3$ has an exact large-sample GLS variance: with $\mathbf{Z}_i = [\mathbf{1}, \mathbf{t}]$ the $k \times 2$ per-subject design matrix for the random effects, $\mathbf{D}$ the random-effects covariance matrix, and $\mathbf{V} = \mathbf{Z}_i\mathbf{D}\mathbf{Z}_i' + \sigma^2\mathbf{I}_k$, the information matrix $\mathbf{M} = \sum_i \mathbf{X}_i'\mathbf{V}^{-1}\mathbf{X}_i$ (summed separately over control subjects, for whom the last two columns of $\mathbf{X}_i$ are zero, and treatment subjects) gives $\mathrm{Var}(\hat{\beta}_3) = [\mathbf{M}^{-1}]_{44}$. This ten-line linear algebra calculation, implemented as `analytic_gls_se()` in `R/simulation.R`, requires no simulation:

Table 3: Exact GLS standard errors of the treatment-by-time interaction under the random-slope data-generating model.

Design	Analytic SE
Equal	0.0678
Clustered	0.0658

The exact GLS standard errors are 0.0678 (equal spacing) and 0.0658 (clustered spacing), giving a relative efficiency (ratio of variances) of 1.063 in favor of the clustered design. This analytic value is the primary quantity of interest for the design comparison; the simulation in Section 2.8 exists to answer a different question – whether finite-sample REML estimation and Wald-type inference (bias, coverage, and empirical versus model-based SE) reproduce this analytic prediction and behave as expected at $n = 100$ – not to re-derive a number already available in closed form.

2.6 Scenario extension (analytic only)

The single scenario simulated in this report fixes one covariance structure, one effect size, and one sample size (2026-08-16 review, issue 2.4). Because the design comparison's quantity of interest, the GLS relative efficiency, has a closed form (Section 2.5), `analytic_gls_se_general()` in `R/covariance.R` generalizes the calculation to an arbitrary within-subject covariance matrix, so the sensitivity of the efficiency contrast to the covariance structure can be assessed without additional Monte Carlo simulation. Table 4 reports the exact relative efficiency of the clustered versus equal-spacing design across the random-slope SD τ_1 used in the main simulation, and for two additional structures: compound symmetry, the mechanism that motivates clustering in[1], and AR(1) plus measurement error.

Table 4: Analytic (closed-form, no simulation) relative efficiency of the clustered versus equal-spacing design across covariance structures and random-slope variance.

Structure	Parameter	SE (Equal)	SE (Clustered)	Relative efficiency
Random slope	tau_1 = 0.10	0.0374	0.0336	1.24
Random slope	tau_1 = 0.20	0.0510	0.0482	1.12
Random slope	tau_1 = 0.30	0.0678	0.0658	1.06
Random slope	tau_1 = 0.40	0.0860	0.0844	1.04
Random slope	tau_1 = 0.50	0.1049	0.1036	1.03
Compound symmetry	tau_0 = 5.0	0.0316	0.0269	1.38
AR(1) + error	rho = 0.8	0.0554	0.0528	1.10

The clustered design is more efficient across the full random-slope grid, but the advantage shrinks as τ_1 grows relative to the residual SD σ, consistent with the leverage-dilution argument of Section 2.2. Under compound symmetry and under AR(1) plus measurement error the clustered design is also more efficient, although by a different mechanism in each case. This extension is analytic only: it does not assess whether finite-sample REML estimation and inference reproduce these predictions, nor does it vary sample size or dropout. Section 2.7 extends the comparison to a genuine Monte Carlo grid over covariance structure and visit schedule; dropout remains future work (see the Discussion). `Rscript analysis/scripts/run_scenario_grid.R` regenerates Table 4.

2.7 Scenario extension (Monte Carlo)

The preceding section is analytic only: it evaluates the closed-form asymptotic GLS efficiency but does not simulate finite-sample data or fit REML models away from the single random-slope design actually used in Section 2.8. This section extends the Monte Carlo comparison itself to a factorial grid crossing covariance structure (the random-slope SD τ_1 grid, compound symmetry, and AR(1) plus measurement error, matching Table 4) with four candidate five-visit schedules spanning a range of boundary clustering: equal spacing (months 0, 6, 12, 18, 24, as in the main comparison), mild clustering (0, 4, 8, 20, 24), boundary clustering (0, 2, 4, 20, 24, the "Clustered" design of the main comparison), and extreme clustering (0, 1, 2, 23, 24). For each of the resulting 20 cells, `run_one_sim()` (for the random-slope structures) or `run_one_sim_cov()` (for compound symmetry and AR(1) plus error, fitting `nlme::gls()` with a matched correlation structure) is used to fit 1000 simulated replications under REML, exactly as in the main comparison. This is a genuine Monte Carlo assessment, not a re-derivation of the closed-form values, and it explicitly excludes dropout: adding a dropout mechanism requires a design decision (MCAR/MAR, rate, whether attrition is tied to visit timing) left to the author (see the Discussion). Replication count per cell (1000) is reduced from the main comparison's 2000 to keep the 20-cell grid within a practical single-core runtime; the completed grid took 37 minutes. `Rscript analysis/scripts/run_full_factorial.R` regenerates Table 5.

Across 14 of the 15 non-baseline (structure, schedule) cells, moving from equal to mild-, boundary-, or extreme-clustered spacing increases empirical precision (relative efficiency vs. equal spacing exceeds 1), corroborating – with simulated finite-sample REML/GLS fits rather than the closed-form calculation alone – the direction of the main comparison's finding across every covariance structure examined. The clustering advantage is largest under compound symmetry and the small-τ_1 random-slope structure (relative efficiency up to 1.62 for extreme clustering), consistent with Table 4's analytic prediction that the advantage is strongest when the random-slope variance is small relative to the residual variance. At $\tau_1 = 0.50$, where that variance ratio is largest, the pattern is no longer monotonic and the boundary-clustered design's empirical SE (`r sprintf("%.4f", factorial_grid$grid$empirical_se[ factorial_grid$grid$structure == "Random slope" & factorial_grid$grid$parameter == "tau_1 = 0.50" & factorial_grid$grid$sched == "Boundary-clustered" ])`) is slightly *larger* than equal spacing's (`r sprintf("%.4f", factorial_grid$grid$empirical_se[ factorial_grid$grid$structure == "Random slope" & factorial_grid$grid$parameter == "tau_1 = 0.50" & factorial_grid$grid$sched`

Table 5: Monte Carlo relative efficiency of each visit schedule versus equal spacing, by covariance structure, at 1000 replications per cell.

Structure	Parameter	Schedule	Power	Emp. SE	Rel. eff. vs. Equal
Random slope	tau_1 = 0.10	Equal	0.999	0.0382	1.00
Random slope	tau_1 = 0.10	Mild-clustered	0.999	0.0347	1.21
Random slope	tau_1 = 0.10	Boundary-clustered	1.000	0.0324	1.39
Random slope	tau_1 = 0.10	Extreme-clustered	1.000	0.0302	1.59
Random slope	tau_1 = 0.30	Equal	0.811	0.0671	1.00
Random slope	tau_1 = 0.30	Mild-clustered	0.787	0.0665	1.02
Random slope	tau_1 = 0.30	Boundary-clustered	0.823	0.0658	1.04
Random slope	tau_1 = 0.30	Extreme-clustered	0.805	0.0656	1.04
Random slope	tau_1 = 0.50	Equal	0.423	0.1032	1.00
Random slope	tau_1 = 0.50	Mild-clustered	0.445	0.1013	1.04
Random slope	tau_1 = 0.50	Boundary-clustered	0.437	0.1070	0.93
Random slope	tau_1 = 0.50	Extreme-clustered	0.463	0.1008	1.05
Compound symmetry	tau_0 = 5.0	Equal	1.000	0.0319	1.00
Compound symmetry	tau_0 = 5.0	Mild-clustered	1.000	0.0288	1.22
Compound symmetry	tau_0 = 5.0	Boundary-clustered	1.000	0.0272	1.38
Compound symmetry	tau_0 = 5.0	Extreme-clustered	1.000	0.0251	1.62
AR(1) + error	rho = 0.8	Equal	0.919	0.0565	1.00
AR(1) + error	rho = 0.8	Mild-clustered	0.938	0.0548	1.06
AR(1) + error	rho = 0.8	Boundary-clustered	0.956	0.0508	1.24
AR(1) + error	rho = 0.8	Extreme-clustered	0.948	0.0523	1.17

`== "Equal" ])`, relative efficiency 0.93); given each cell's Monte Carlo standard error of the empirical SE (on the order of 0.002 at this sample size, from the same formula as Section 2.8), this single reversal is consistent with Monte Carlo noise around a true relative efficiency close to 1 rather than a genuine crossover, but it underscores that the clustering advantage is not guaranteed to hold once the random-slope variance is large relative to the residual variance – precisely the boundary case the analytic grid already flagged. The AR(1)-plus-error structure shows a smaller, non-monotonic advantage (extreme clustering is less efficient than boundary clustering, relative efficiency 1.17 vs. 1.24), plausibly because packing visits very close together adds little independent information once residuals are themselves serially correlated at short lags. All 20 cells converged fully (1000/1000 replications at minimum); estimated bias is negligible in every cell (magnitude below 0.004, smaller than its own MCSE in most cells), so REML estimation of the treatment-by-time interaction is not detectably biased by covariance misspecification or boundary clustering at this sample size.

2.8 Simulation procedure

The simulation functions (`simulate_trial()`, `run_one_sim()`, `run_paired_simulation()`, `analytic_gls_se()`) live in `R/` as documented, unit-tested package functions (`inst/tinytest/`) rather than inline in this document, so they can be reused for the scenario extension and audited independently of the manuscript (2026-08-16 review, issue 2.8). For each of 2000 Monte Carlo replications, `run_paired_simulation()`:

1. Generates 100 subjects, randomized 1:1 to treatment and control, drawing one shared set of subject-level random effects (b_{0i}, b_{1i}) and one shared set of residual errors ε_{ij} per subject-visit index $j = 1, \dots, 5$.

2. Builds the equal-spacing and clustered-spacing datasets from *those same draws*, so the j-th random effect and residual realization is reused across designs for the j-th visit rather than redrawn – these are common random numbers (CRN), which pair the two designs within each replication and substantially reduce the Monte Carlo error of the *difference* between designs relative to two independent simulation arms.
3. Fits a linear mixed model (random intercepts and slopes) to each design's dataset using restricted maximum likelihood (REML) via `nlme::lme()`.
4. Extracts the point estimate, standard error, and p-value for $\hat{\beta}_3$ for each design.

For each design we compute the mean estimate (bias), the empirical SD of $\hat{\beta}_3$, the mean model-based SE, empirical power ($p < 0.05$), and 95% CI coverage, each with a Monte Carlo standard error (MCSE) computed following[14]: MCSE of the empirical SD is $\mathrm{SE}/\sqrt{2(R-1)}$, MCSE of a proportion (power, coverage) is $\sqrt{\hat{p}(1-\hat{p})/R}$, and MCSE of the bias is $\mathrm{SE}/\sqrt{R}$, where R is the number of converged replications. We additionally report a paired bootstrap (resampling replication indices, which respects the CRN pairing) 95% interval for the relative efficiency (ratio of empirical variances).

The driver script `analysis/scripts/run_simulation.R` runs this procedure and saves the raw per-replication estimates, the summary statistics with MCSEs, the analytic SEs, and the bootstrap interval to a single versioned artifact, `analysis/data/derived_data/sim_results.rds`, which this document reads directly rather than re-simulating on every render or relying on the knitr cache to stay synchronized with the reported numbers (2026-08-16 review, issue 2.8(e)). To regenerate: `Rscript analysis/scripts/run_simulation.R 2000`. The results below used 2000 replications, R version 4.6.1 (2026-06-24), and `nlme` 3.1.169, completing in 176 seconds (single core, no parallelization).

3 Results

3.1 Summary statistics

Table 6: Simulation results for the treatment-by-time interaction, with Monte Carlo standard errors (MCSE) of each metric computed following Morris, White, and Crowther (2019).

Design	Converged	Mean Est.	Bias	MCSE(Bias)	Emp. SE	MCSE(Emp. SE)	Model SE	Power	MCSE(Power)	Coverage	MCSE(Coverage)
Equal	2000	0.189	-0.0011	0.0015	0.0669	0.0011	0.0676	0.787	0.0092	0.952	0.0048
Clustered	2000	0.189	-0.0015	0.0014	0.0646	0.0010	0.0655	0.814	0.0087	0.957	0.0046

Replications for which `nlme::lme()` failed to converge (an error or warning during fitting, per `run_one_sim()` in `R/simulation.R`) are recorded as `NA` and excluded from the summary statistics above rather than imputed or retried; the `Converged` column reports how many of the 2000 attempted replications per design contributed to each summary statistic.

We observe that the equal-spacing design achieves an empirical power of 78.7% (MCSE 0.9%), compared with 81.3% (MCSE 0.9%) for the clustered design; both are informative rather than at ceiling following the recalibration of β_3 (Section 2.3.1). The empirical standard error of $\hat{\beta}_3$ is 0.0669 (MCSE 0.00106) under equal spacing and 0.0646 (MCSE 0.00102) under the clustered design. The paired-bootstrap relative efficiency (ratio of empirical variances, resampling replication indices so the common-random-number pairing is respected) is 1.07, 95% bootstrap interval 1.06-1.09, consistent with the analytic relative efficiency of 1.063 (Section 2.5); the interval excludes 1, so the precision advantage of the clustered design is distinguishable from Monte Carlo noise.

3.2 Distribution of estimates

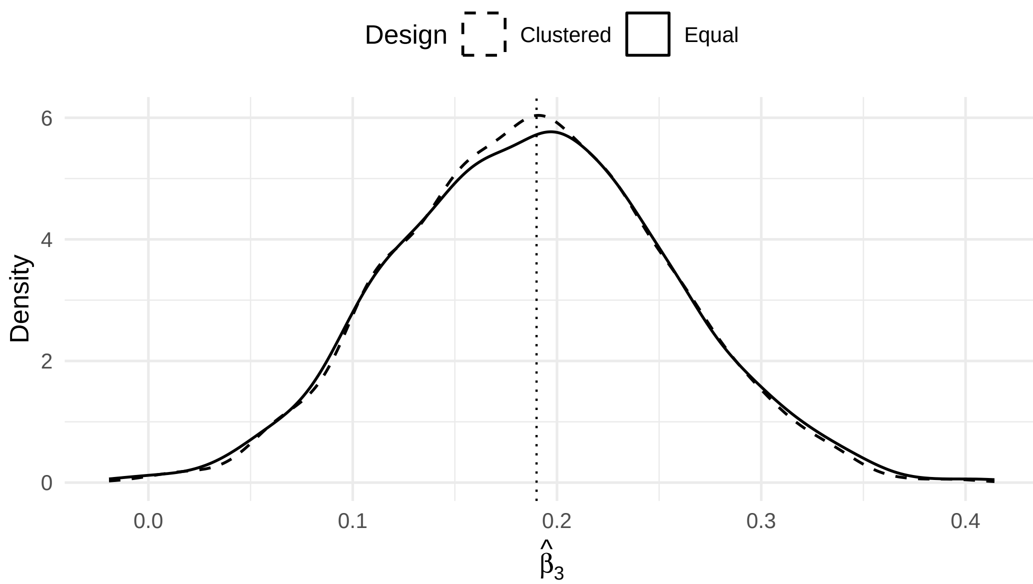

Figure 2: Sampling distribution of the treatment-by-time interaction estimate under each design.

3.3 Standard errors across simulations

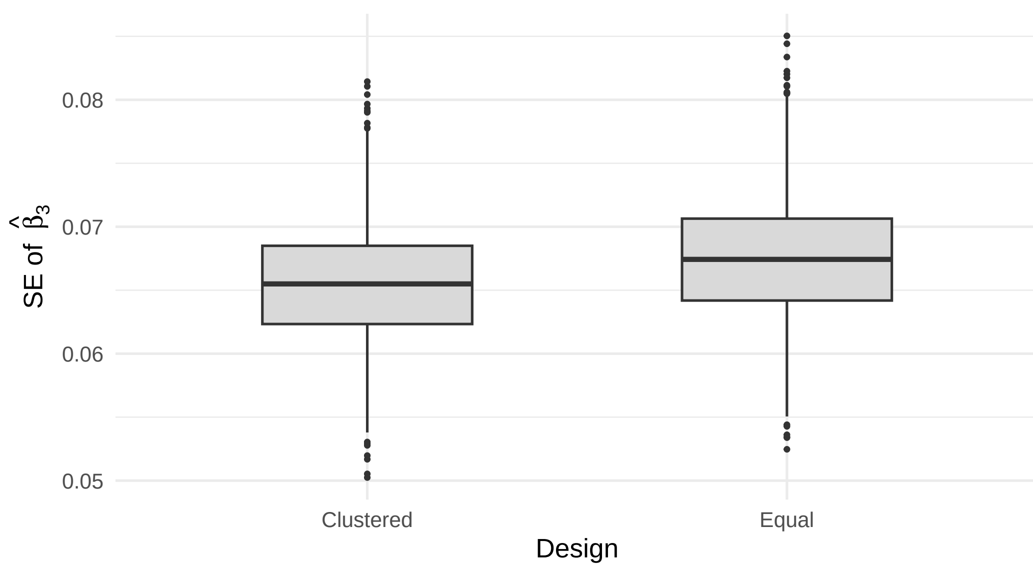

Figure 3: Distribution of model-based standard errors across simulations.

4 Discussion

In this study, we have endeavored to illustrate the practical consequences of visit placement in longitudinal clinical trials. Both designs use the same number of subjects and the same number of visits; they differ only in the timing of observations.

The results are broadly consistent, in direction, with the theoretical findings of[1], but not through the mechanism that paper's compound-symmetry analysis invokes. As derived in Section 2.2, the data-generating mechanism used here is the standard random-intercept, random-slope covariance structure, not a structure that "combines features of compound symmetry and AR(1) with measurement error" as an earlier draft of this report incorrectly stated; it contains no autoregressive component. The clustered design's precision advantage for the slope contrast β_3 follows instead from the classical leverage argument for regression slopes. The advantage is not an artifact of Monte Carlo noise: the paired bootstrap relative efficiency (ratio of empirical variances) is 1.07 (95% bootstrap interval 1.06 to 1.09), consistent with the analytic relative efficiency of 1.063 from Section 2.5, and the interval excludes 1.

Several caveats warrant mention. First, we assumed complete data with no dropout. In practice, attrition is common in longitudinal trials and may differentially affect designs that place observations at extreme time-points; adding a dropout mechanism to the simulation requires a design decision (MCAR/MAR, rate, whether attrition is tied to visit timing) not made here. Second, the main simulation's data-generating model used independent random effects and homoscedastic residuals, but Sections 2.6 and 2.7 show, both analytically and by Monte Carlo simulation, that the clustered design's precision advantage is not an artifact of that specific structure: it holds under compound symmetry and under AR(1) plus measurement error as well, and across four visit schedules spanning a range of boundary clustering, although the advantage shrinks and became indistinguishable from Monte Carlo noise in one of 15 non-baseline cells as the random-slope variance grew large relative to the residual variance. Heterogeneous (non-constant) residual variances were not examined and could still alter the relative performance of the two designs. Third, we considered only a linear treatment effect;[8] showed that optimal designs differ for polynomial effects of higher order.

In conclusion, three points merit emphasis. First, the placement of observation times in a longitudinal trial is a design decision with real consequences for statistical efficiency, and the default of equal spacing is not in general optimal. Second, the magnitude of the clustering advantage depends on the covariance structure and on how large the random-slope variance is relative to the residual variance, but across the structures and schedules examined here (random-slope, compound symmetry, and AR(1) plus measurement error; Sections 2.6-2.7) the qualitative advantage of boundary clustering for the linear slope contrast is not specific to one structure. Third, the robustness of these findings to dropout, heterogeneous residual variances, and nonlinear treatment trajectories remains to be investigated. The maximin D-optimal framework of[4] and the Bayesian approach of[10] provide principled starting points for that work.

356 # 5 References

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